

## TATARSK, Russian Federation

WMO: 296050

Lat: 55.224N

Lon: 76.019E

Elev: 368

StdP: 14.50

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -29.8      | -24.7 | -35.0                         | 0.8 | -29.4 | -29.8 | 1.0 | -24.4 | 17.9                  | 11.1 | 15.9 | 14.1 | 2.7                      | 20   | 0.610 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.3                   | 86.0            | 65.2 | 82.8 | 64.5 | 79.9 | 63.3 | 68.5                | 79.9 | 66.9 | 77.8 | 65.4 | 75.6 | 7.7                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 64.9                            | 93.5 | 72.6 | 63.1 | 87.7 | 71.2 | 61.4 | 82.5 | 70.2 | 33.0          | 79.7 | 31.8 | 78.3 | 30.6 | 75.8 | 77.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 17.0          | 14.7 | 13.4 | DB                         | -34.0 | 91.5               | 8.1 | 3.0                                                | -39.8 | 93.7       | -44.6 | 95.5       | -49.1 | 97.2       | -55.0 | 99.4 | (4) |
| (5)               |      |      | WB                         | -34.1 | 71.6               | 8.1 | 2.1                                                | -40.0 | 73.1       | -44.7 | 74.3       | -49.2 | 75.4       | -55.1 | 77.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |               | Annual  | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|---------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |               | (d)     | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg         | 35.4    | -0.8  | 4.4   | 19.8  | 39.6  | 53.7  | 63.5  | 66.5  | 61.8  | 51.1  | 37.9  | 20.0  | 5.2   |  |  |
|                                                                        | DBStd         | 25.91   | 14.15 | 12.75 | 10.59 | 10.54 | 9.80  | 7.71  | 6.31  | 6.65  | 8.26  | 9.02  | 12.83 | 14.06 |  |  |
|                                                                        | HDD50         | 6972    | 1575  | 1277  | 937   | 342   | 76    | 5     | 0     | 1     | 85    | 385   | 900   | 1389  |  |  |
|                                                                        | HDD65         | 11057   | 2040  | 1697  | 1402  | 763   | 367   | 117   | 59    | 145   | 423   | 840   | 1350  | 1854  |  |  |
|                                                                        | CDD50         | 1637    | 0     | 0     | 0     | 29    | 191   | 411   | 512   | 366   | 117   | 11    | 0     | 0     |  |  |
|                                                                        | CDD65         | 248     | 0     | 0     | 0     | 0     | 18    | 73    | 107   | 45    | 5     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74         | 2486    | 0     | 0     | 0     | 7     | 286   | 741   | 920   | 448   | 84    | 0     | 0     | 0     |  |  |
|                                                                        | CDH80         | 641     | 0     | 0     | 0     | 0     | 68    | 206   | 245   | 108   | 14    | 0     | 0     | 0     |  |  |
|                                                                        | Wind          | WSAvg   | 6.5   | 5.8   | 6.3   | 7.1   | 7.5   | 7.1   | 6.1   | 5.7   | 6.1   | 6.6   | 6.9   | 6.5   |  |  |
|                                                                        | Precipitation | PrecAvg | 14.8  | 0.6   | 0.5   | 0.5   | 0.8   | 1.3   | 2.0   | 2.6   | 2.0   | 1.5   | 1.1   | 0.8   |  |  |
|                                                                        |               | PrecMax | 19.3  | 1.3   | 1.3   | 1.6   | 2.6   | 3.0   | 4.6   | 6.9   | 5.4   | 4.4   | 2.8   | 2.7   |  |  |
|                                                                        |               | PrecMin | 10.1  | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.2   | 0.6   | 0.0   | 0.2   | 0.0   |  |  |
|                                                                        |               | PrecStd | 2.7   | 0.3   | 0.3   | 0.4   | 0.6   | 0.9   | 1.1   | 1.7   | 1.1   | 1.0   | 0.6   | 0.6   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%          | DB      | 32.6  | 33.7  | 44.3  | 73.6  | 86.1  | 89.3  | 90.0  | 87.9  | 81.5  | 66.8  | 44.9  | 34.6  |  |  |
|                                                                        |               | MCWB    | 31.4  | 31.5  | 37.9  | 54.9  | 61.3  | 65.4  | 67.8  | 66.1  | 59.8  | 51.5  | 41.9  | 32.7  |  |  |
|                                                                        | 2%            | DB      | 28.3  | 30.7  | 39.1  | 67.0  | 81.0  | 85.5  | 85.7  | 82.8  | 75.7  | 61.0  | 41.4  | 32.1  |  |  |
|                                                                        |               | MCWB    | 27.2  | 29.2  | 35.0  | 52.6  | 59.6  | 64.5  | 66.0  | 65.3  | 58.7  | 49.2  | 39.2  | 30.9  |  |  |
|                                                                        | 5%            | DB      | 24.6  | 27.1  | 36.2  | 61.6  | 76.8  | 81.9  | 82.4  | 79.1  | 70.2  | 55.8  | 37.7  | 28.4  |  |  |
|                                                                        |               | MCWB    | 23.3  | 25.6  | 33.3  | 49.9  | 57.9  | 64.1  | 65.7  | 63.7  | 56.5  | 46.9  | 35.4  | 27.4  |  |  |
|                                                                        | 10%           | DB      | 20.6  | 23.0  | 34.0  | 56.3  | 71.6  | 78.2  | 79.3  | 75.1  | 65.3  | 51.1  | 34.4  | 24.1  |  |  |
|                                                                        |               | MCWB    | 19.6  | 21.7  | 31.6  | 46.8  | 55.6  | 62.7  | 64.8  | 62.1  | 54.0  | 44.8  | 32.6  | 22.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%          | WB      | 31.5  | 32.0  | 38.3  | 56.9  | 64.7  | 70.6  | 71.0  | 69.0  | 63.0  | 53.6  | 42.9  | 33.2  |  |  |
|                                                                        |               | MCDB    | 32.4  | 33.1  | 43.3  | 69.3  | 79.3  | 82.3  | 82.7  | 80.5  | 76.7  | 61.9  | 44.5  | 34.0  |  |  |
|                                                                        | 2%            | WB      | 27.2  | 29.0  | 35.6  | 53.9  | 60.9  | 67.5  | 69.0  | 67.0  | 59.9  | 50.9  | 39.1  | 30.9  |  |  |
|                                                                        |               | MCDB    | 28.3  | 30.6  | 38.3  | 64.5  | 76.9  | 79.0  | 80.3  | 78.5  | 71.7  | 58.3  | 41.5  | 32.2  |  |  |
|                                                                        | 5%            | WB      | 23.4  | 25.7  | 33.7  | 50.5  | 58.6  | 65.7  | 67.5  | 65.4  | 57.5  | 47.8  | 35.6  | 27.3  |  |  |
|                                                                        |               | MCDB    | 24.5  | 27.1  | 36.0  | 60.8  | 74.4  | 76.8  | 78.1  | 75.5  | 68.3  | 54.7  | 37.5  | 28.4  |  |  |
|                                                                        | 10%           | WB      | 19.6  | 21.8  | 31.7  | 47.3  | 56.4  | 64.0  | 66.1  | 63.5  | 54.9  | 45.0  | 33.0  | 22.9  |  |  |
|                                                                        |               | MCDB    | 20.6  | 23.0  | 34.0  | 55.8  | 69.8  | 75.3  | 76.0  | 72.9  | 64.3  | 50.8  | 34.3  | 24.0  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB         | MDBR    | 13.8  | 15.0  | 15.2  | 16.6  | 21.0  | 19.5  | 18.3  | 19.0  | 18.1  | 13.7  | 11.1  | 13.3  |  |  |
|                                                                        |               | MCDBR   | 16.2  | 13.4  | 13.3  | 25.1  | 28.4  | 25.3  | 23.4  | 25.8  | 26.7  | 23.0  | 11.0  | 15.3  |  |  |
|                                                                        |               | MCWBR   | 15.7  | 12.8  | 10.4  | 13.9  | 12.1  | 9.6   | 8.3   | 10.7  | 12.4  | 13.3  | 9.4   | 15.0  |  |  |
|                                                                        |               | MCDBR   | 15.8  | 13.2  | 12.4  | 23.9  | 25.2  | 21.4  | 19.8  | 21.4  | 24.3  | 20.2  | 11.2  | 15.1  |  |  |
|                                                                        | 5% WB         | MCWBR   | 15.3  | 12.7  | 10.4  | 13.9  | 11.8  | 9.8   | 8.5   | 9.9   | 12.2  | 12.5  | 10.0  | 14.8  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub          |         | 0.283 | 0.315 | 0.374 | 0.405 | 0.400 | 0.410 | 0.430 | 0.422 | 0.365 | 0.353 | 0.322 | 0.286 |  |  |
|                                                                        | taud          |         | 2.039 | 2.010 | 1.923 | 2.118 | 2.230 | 2.282 | 2.224 | 2.215 | 2.333 | 2.227 | 2.087 | 1.990 |  |  |
|                                                                        | Ebn at Noon   |         | 192   | 230   | 242   | 257   | 267   | 265   | 256   | 250   | 250   | 218   | 173   | 155   |  |  |
|                                                                        | Edh at Noon   |         | 22    | 32    | 45    | 43    | 41    | 39    | 41    | 39    | 30    | 26    | 21    | 18    |  |  |
| (44) All-Sky Solar Radiation                                           |               | RadAvg  | 304   | 648   | 1152  | 1472  | 1747  | 1890  | 1732  | 1397  | 989   | 498   | 239   | 197   |  |  |
| (45)                                                                   |               | RadStd  | 22    | 31    | 69    | 120   | 159   | 149   | 116   | 126   | 94    | 60    | 38    | 22    |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (12 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page